

ROMA URBE, Italy

WMO: 162350

Lat: 41.952N

Lon: 12.501E

Elev: 24

StdP: 101.04

Time Zone: 1.00 (EUC)

Period: 00-11

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.4	-1.1	-9.2	1.7	1.5	-7.4	2.0	2.6	8.6	8.4	7.7	8.3	0.5	350	0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	34.8	23.1	33.1	22.6	31.9	22.3	25.1	31.1	24.1	30.2	23.3	29.2	3.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.0	28.0	22.2	17.0	27.2	21.3	16.0	26.4	76.7	31.1	72.7	30.1	69.4	29.2	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.4	5.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	7.4	8.2	10.9	13.9	18.0	22.1	24.9	21.1	17.4	12.4	8.7
		DBStd	6.74	2.72	2.76	2.82	2.18	2.46	3.19	2.10	2.17	3.04	2.46	3.16
		HDD10.0	252	90	63	24	1	0	0	0	0	0	12	61
		HDD18.3	1564	340	284	231	132	36	7	0	12	46	179	297
		CDD10.0	2394	8	13	52	119	248	363	463	332	230	85	23
		CDD18.3	664	0	0	0	0	26	119	204	201	94	18	2
		CDH23.3	6799	0	0	1	12	284	1214	2285	2098	772	131	3
		CDH26.7	2801	0	0	0	0	59	474	1079	951	228	11	0
(14)	Wind	WSAvg	1.9	1.6	2.0	2.2	1.9	2.0	2.1	2.4	2.3	2.0	1.4	1.7
(15)	Precipitation	PrecAvg	842	88	70	73	64	49	42	16	38	81	113	128
		PrecMax	1189	160	129	172	204	130	113	57	171	322	329	307
		PrecMin	569	2	4	15	13	4	0	0	0	0	21	19
		PrecStd	161	42	38	46	47	29	31	18	42	73	96	82
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.0	18.1	20.8	24.9	30.2	35.0	36.1	36.9	32.2	28.0	23.2
			MCWB	13.0	11.4	14.2	16.7	19.9	23.1	24.2	23.6	21.9	20.1	17.5
		2%	DB	15.2	16.9	19.2	22.9	28.1	33.1	34.8	34.8	30.6	26.2	21.1
			MCWB	11.8	11.1	13.2	15.8	19.1	22.3	23.1	23.3	20.9	19.5	16.9
		5%	DB	14.0	15.2	18.1	21.2	26.2	31.6	33.2	32.9	29.1	24.9	19.8
			MCWB	10.7	10.5	12.7	14.6	18.3	21.8	22.5	23.0	20.2	18.9	15.8
		10%	DB	12.8	13.9	17.0	19.9	24.8	29.9	32.0	31.2	27.9	23.3	18.2
			MCWB	9.7	9.8	12.2	13.9	17.6	21.2	22.3	22.2	20.2	18.0	14.7
		0.4%	WB	14.6	13.5	15.8	17.7	21.2	24.7	26.6	25.9	24.7	22.6	19.1
			MCDB	15.8	16.1	19.2	23.4	27.1	31.1	32.2	31.9	29.2	25.3	21.4
(28)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.8	12.1	14.6	16.2	20.2	23.7	25.4	24.9	23.1	20.9	17.7
			MCDB	14.7	14.8	17.8	21.3	26.1	30.3	31.3	31.0	27.4	24.4	20.1
		2%	WB	11.3	11.2	13.7	15.4	19.2	22.7	24.2	24.1	22.0	19.7	16.5
			MCDB	13.5	14.6	16.8	20.1	25.0	29.5	30.4	30.2	26.9	23.5	19.1
		5%	WB	10.0	10.1	12.8	14.4	18.2	21.7	23.2	23.3	21.0	18.6	15.3
			MCDB	12.1	13.5	15.9	18.8	23.6	28.2	29.4	29.1	25.9	22.4	17.8
		10%	WB	10.0	10.1	12.8	14.4	18.2	21.7	23.2	23.3	21.0	18.6	15.3
			MCDB	12.1	13.5	15.9	18.8	23.6	28.2	29.4	29.1	25.9	22.4	17.8
		0.4%	WB	14.6	13.5	15.8	17.7	21.2	24.7	26.6	25.9	24.7	22.6	19.1
			MCDB	15.8	16.1	19.2	23.4	27.1	31.1	32.2	31.9	29.2	25.3	21.4
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.7	10.5	10.6	11.6	11.9	12.5	13.5	13.1	12.0	11.4	9.8
			MCDBR	11.0	11.9	12.8	13.6	14.7	14.5	15.6	14.9	13.9	13.3	11.4
		5% WB	MCWBR	8.5	7.7	8.1	8.1	7.1	5.8	6.5	6.2	6.4	7.5	7.2
			MCWBR	9.6	9.4	11.1	12.6	13.6	13.3	13.5	13.2	11.5	11.2	10.1
				8.5	6.9	7.5	8.0	7.1	6.4	7.0	6.6	6.3	7.2	7.5
(40)	Clear-Sky Solar Irradiance	taub	0.349	0.374	0.415	0.444	0.449	0.442	0.433	0.432	0.424	0.411	0.384	0.350
		taud	2.409	2.314	2.195	2.133	2.166	2.241	2.272	2.285	2.302	2.306	2.313	2.403
		Ebn at Noon	785	819	823	826	831	836	840	829	807	766	728	747
		Edh at Noon	81	103	130	148	147	137	132	126	116	103	88	75
(44)	All-Sky Solar Radiation	RadAvg	1.74	2.64	3.87	5.15	6.31	7.19	7.38	6.44	4.70	3.20	1.96	1.56
		RadStd	0.17	0.27	0.45	0.38	0.54	0.31	0.34	0.35	0.23	0.28	0.18	0.17

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	+0.40	N/A	N/A	+0.51	N/A	N/A	N/A	-78	+135	+69

Nomenclature: See separate page